ROYAL CANADIAN NAVY QUALIFICATION STANDARD AND PLAN (QSP)

Maritime Surface and Sub-Surface (MARS) (00207),
Naval Combat Information Operator (NCIOP) (00114),
Naval Communicator (NAV COMM) (00299),
Sonar Operator (SONAR OP) (00324), and
Weapons Engineering Technician (W ENG TECH) (00366)

Submarine Second Officer of the Watch (2OOW)

NQual – ALDE

MITE Course IDs:

Residential Phase – 120231

OJPR – 120232

RECORD OF AMENDMENTS

- This QSP is developed under the authority of the Royal Canadian Navy (RCN) through its Training Authority (TA), Director Naval Personnel and Training (D Nav P&T) and the Curriculum Control Authority (CCA), Commandant of Canadian Forces Naval Operations School (CFNOS). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the change(s) must be amplified in Annex J "Course History".

REVISION	APPROVAL DATE	AUTHORITY
1.0	4 Apr 2014	Commandant CFNOS
1.0	16 Dec 2015	NPTG Quality Control

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FOREWORD

1. This training document is developed under the authority of the Chief of the Naval Staff (C Naval) and in accordance with the following references:

a. A-P9-050 Series, Canadian Forces Manual of Individual Training and Education;

b. A-PD-055/PP-001 Occupation Specification (Maritime Surface and Sub Surface) dated 15 Dec 05;

c. A-PD-055/PP-002 Occupation Specification (SONAR Op) dated 8 Nov 11;

d. A-PD-055/PP-002 Occupation Specification (W Eng Tech) dated 1 Sep 11;

e. A-PD-055/PP-002 Occupation Specification (NAV COMM) dated 28 Jul 05; and

f. A-PD-055/PP-002 Occupation Specification (NCI Op) dated 1 Sep 12.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to the CCA for approval.

5. The designated CCA(s) for this training is indicated below:

CCA	PHASE / MODULE	MITE COURSE ID	NQUAL
Commandant CFNOS	All	120231/120232	ALDE

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT CFNOS

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training to perform the duties of the Second Officer of the Watch (2OOW) aboard HMC Submarines. This QSP is designed to assist training establishments in developing, conducting, evaluating and validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with MARCORD 9-30 – Maritime Command Course Scheduling. For this course, a period of instruction is 50 minutes in duration.

	Periods
Actual Training Time (ATT) (Sum of all EOs/POs excluding PC/ECs)	382.0
Enabling Checks (ECs)	38.0
Performance Checks (PCs)	27.0
Base Administrative Time (BAT)	8.0 (1 day)
Training Travel Time (TTT)	0
Sub-total	460.0 **57.5 (58) trg days**
Training Administrative Time (TAT) / Physical Training (PT) * **1 period per training day is allocated IAW MARCORD 9-30** * **min of 2-3 PT periods per week IAW MARGEN 009-09** * **partial periods must be rounded up / no part periods for TAT/PT**	58.0
Course Duration Calculation (ATT+PCs/ECs+BAT+TTT+TAT/PT) **8 periods** * **Partial days must be rounded up**	522.0 **66 trg days**

RESOURCES

3. Identify key personnel and facilities required to conduct the training. Additional training establishment resources can be listed in Annex H if required.

REQUIRED COURSE STAFFING				
POSITION	**RANK**	**QUALIFICATIONS REQUIRED**	**QTY**	**PO/ EO**
Course Officer	Lt(N)	AILS, AILV and AILT	1	
Div CPO	CPO2	AILS	1	

Senior Instructor	Lt(N)	AILS, AILV and AILT	1	
Technical Instructor	Lt(N) / PO1	AILS, AILV and AISF or AISE	1	HOD qualified NTO or MARENG Cert 3.
Trainer Support Personnel	OS / Lt(N)	1 x AILS, AIND – SCC 1 x AILS – Helmsman	2	Must act as Helmsman and SCC1 Watchkeeper

CRITICAL TRAINING RESOURCES	NO. OF DAYS
Sherwood Command Team Trainer	22
Ship Control Trainer	19
CFNOS Voice Procedures Trainer	4
Classroom	22

COURSE CAPACITY

4. Planning for the conduct of training is based on the MARCOM Production Books (Gold, Silver and Blue) following guidelines in MARCORD 9-30 - Maritime Command Course Scheduling:

a. Maximum – 8;

b. Minimum – 4; and

c. Optimum – 8.

MITE

5. Loading. The authority to load members onto this course lies with the Training Establishment (TE). D Nav P & T is to be included as an INFO addressee on all loading messages.

6. Scheduling. Each session of this course shall be scheduled IAW MARCORD 9-47 - Maritime Command Individual Training and Education Policy and MARCORD 9-30 - Maritime Command Course Scheduling in MITE by the Training Establishment conducting the training. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays.

7. Prerequisites. To be selected for this course the loading authority must ensure that trainees have the following prerequisites:

a. Hold the rank of Petty Officer 2nd class or higher;

b. Minimum Level II Security Clearance;

c. Basic Submarine Qualification (NQual AILS, MITE 109201, 109381, 109285);

d. For NCMs:

(1) Submarine Control Room Watchkeeper (Dived) qualified (NQual AIMZ, MITE 109241);

(2) Submarine Control Room Watch Supervisor qualified (NQual AILR, MITE 109336);

e. For SONAR Ops, possess the following qualifications:

(1) Sonar Operator Submarines (QL4) (NQual AILO, MITE 109272, 109273);

(2) Sonar Control Supervisor (NQual AIXJ, MITE 115043); and

(3) Submarine Sonar Underwater Controller 1 (NQual AILE, MITE 109344); or

f. For NAV COMMs, possess the following qualifications:

(1) Communications Information Systems Network Supervisor (QL6) (NQual AIXD, MITE 114950); and

(2) Submarine Naval Communicator (NQual AIIW, MITE 109270, 109271); or

g. For NCIOPs, possess the following qualifications:

(1) Submarine Naval Combat Information Operator (NQual AILD, MITE 109268, 109269); and

(2) NCIOP Operations Room Supervisor (QL6A) (NQual AAXA, MITE 108906).

8. Qualification. Successful completion of this course constitutes the primary means of attaining the Second Officer of the Watch (NQual ALDE). Additionally, the following MITE Course IDs will be granted:

a. Second Officer of the Watch (Residential) – MITE Course ID (120231); and

b. Second Officer of the Watch (OJPR) – MITE Course ID (120232).

COURSE REPORTING

9. Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 - Course Reporting and Certificates and DAOD 5031-9 - Course Reporting and Certificates.

EVALUATION

10. Program evaluation shall be conducted IAW the guidance provided in A-P9-050-000 CFITES Vol. 11 - Evaluation and Validation Techniques.

VALIDATION

11. This training will be validated every five years CFITES Vol. 8 - Validation of Instructional Programs and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

12. PLARs shall be conducted in accordance with MARCORD 9-47 - Maritime Command Individual Training and Education Policy, DAOD 5031-1 - Canadian Forces Military Equivalencies Program and CFITES Vol 12 - Canadian Forces Military Equivalencies Programme: Prior Learning Assessment.

SPECIAL INSTRUCTIONS

13. CFNOS to report trainee progress to DCSF Sustainment. Upon course completion, trainee OJT progress is to be tracked by their home units and managed by CFNOS Submarine QA.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT CFNOS

PO 001 Supervise Ship Control (SCOOW)

1. Performance. (T0009) Supervise Ship Control (SCOOW)

2. Conditions

   a. Given:

      (1) Submarine;

      (2) Supervision; and

      (3) References.

   b. Denied: Nil.

   c. Environmental:

      (1) At sea; and

      (2) Watch Dived.

3. Standard. The SCOOW shall, IAW references (T0009) supervise ship control by:

   a. (T0008) Maintaining Trim;

   b. (T0012) Employing Manoeuvring Limitation Diagram (MLD);

   c. (T0006) Conducting dived Emergency Operating Procedures (EOPs) as SCOOW; and

   d. (T0007) Conducting dived SOPs.

4. References. A1, A2, A3, A4, A7, A8, B5, B6, B7, B8, D2, D9, D10, D13, D16, D21, D22, D23, D24

5. Task Numbers. T0006, T0007, T0008, T0009, T0012

6. Training Limitations. Training will be conducted in SCT/MCT simulators.

PO 002 Conduct Dived Operations

1. Performance. (T0011) Conduct Dived Operations.

2. Conditions

   a. Given:

      (1) Submarine;

      (2) Supervision; and

      (3) References.

   b. Denied: Nil.

   c. Environmental:

      (1) At sea; and

      (2) Watch Dived.

3. Standard. The Second Officer of the Watch (2OOW) shall (T0011) conduct dived operations by:

   a. (T0002) Interpreting Submarine sensor data;

   b. (T0003) Managing Submarine fire control system, to include:

      (1) Conducting Broadband Target Motion Analysis; and

      (2) Conducting Narrowband Target Motion Analysis.

   c. (T0001) Using external communication systems;

   d. (T0010) Coordinating the Watch; and

   e. (T0005) Calculating Submarine position.

4. References. A5, A6, B1, B2, B3, B4, B9, B10, B11, B12, B13, B14, B16, B17, B18, B19, B21, D1, D2, D3, D4, D5, D6, D7, D8, D9, D11, D12, D13, D14, D15, D16, D17, D19, D20, D21, D22, D24, D25

5. Task Numbers. T0001, T0002, T0003, T0005, T0007, T0010, T0011

6. Training Limitations. Training will be conducted in the Sherwood Command Team Trainer (SCTT)

PO 003 Conduct Periscope Watchkeeping

1. Performance. (T0004) Conduct Periscope Watchkeeping

2. Conditions

   a. Given:

      (1) Submarine;

      (2) Supervision; and

      (3) Stop watch.

   b. Denied: Nil.

   c. Environmental:

      (1) At sea; and

      (2) Watch Dived.

3. Standard. The 2OOW shall, IAW references, (T0004) conduct periscope watchkeeping by maintaining submarine safety at periscope depth.

4. References. B3, B15, B20, D1, D2, D12, D15, D18, D19, D21, D23.

5. Task Numbers. T0004, T0006

6. Training Limitations. Training shall be conducted in the SCTT.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT CFNOS

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification. PCs will be administered in the form of (PRACTICAL or WRITTEN) examinations and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

2. Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

3. Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

4. Assessment of trainees is to be done IAW CFITES Vol. 7 – Evaluation.

5. Standards provides quality control by designing, administering and scoring all Performance Checks. Any PC not administered or invigilated by standards staff will be administered by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the Instructional Staff.

PROGRESS MONITORING

6. Monitoring learning is the responsibility of all instructional and standards staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes and TE SOs or SOPs.

7. Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the TRB or PRB (if required); and

d. Information to assist in completion of the CF 377 Course Report.

8. Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW Training Establishment SOs or SOPs.

9. PC failures and supplemental PC failures are to be handled IAW Training Establishment SOs or SOPs. Supplemental ECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their EC or PC failure.

TRAINING REVIEW BOARD

10. TRBs shall be convened and conducted IAW with MARCORD 9-60 - Training Review Board Process.

COURSE REPORTS

11. Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12, NTI 014/2010 or applicable Training Establishment training directives.

12. Trainee will be graded pass or fail. IAW NTI 1/08 Course Reporting and Certificates, trainees will not be ranked.

13. Where necessary for Rank Qualifications, the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and / or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the training establishment.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDANT CFNOS

PO 001 SUPERVISE SHIP CONTROL

EO 001.01 Maintain Trim

1. Performance. (T0008) Maintain Trim

2. Conditions

   a. Given:

      (1) Classroom;

      (2) SCT/MCT; and

      (3) References.

   b. Denied: Nil

   c. Environmental: Classroom.

3. Standard. The trainee shall, IAW references, maintain:

   a. Ordered depth; and

   b. 1 deg bow down angle.

4. Time Required

   a. EO - 18 x 50 min periods;

   b. EC - 2 x 50 min periods; and

   c. PC - 16 x 50 min periods (2 per trainee).

5. Instructional Methods

   a. Interactive lecture - 6 x 50 min periods; and

   b. Demonstration/practice - 12 x 50 min periods; and

   c. OJT.

6. Training Aids. SCANDEX.

7. Test Details. This EO will be tested in EC 001.01 and PC 001.

8. References. A8, B5, D2, D13, D16, D24.

9. Remarks. Two support personnel required in the SCT for Helmsman and SCC 1 Watchkeeper.

10. Teaching Points

	Description	Method	Time	Reference

TP1	Submarine Trim to include:
	• Principles of Trimming;
	• Catching a Trim;
	• Determining if SM is in Trim;
	• Catching a Stopped Trim;
	• Trimming Techniques;
	• Utilise OMC Indication; and
	• Describe OMC Operation:
	  • Architechture,
	  • Modes of Operation
	  • Alarms
	  • Available Instrumentation; and
	  • Interpretation of readings.
		Interactive Lecture/ Practical	3 periods/ 7 practical	D24, B5, D2

TP2	(T0008) Maintaining Trim, to include: • Snorting; • Domestics; and • Ditching Gash.	Practical	5 periods	D24, D2, D13
TP3	Effects of bulk liquids on trim.	Interactive Lecture	1 period	D24
TP4	Depth Change Compensations.	Interactive Lecture	1 period	D24, A8
TP5	Bathymetry ▪ Equipment ▪ Bathythermogram interpretation ▪ Impact on trim	Lecture	1 period	D16, D24

ENABLING CHECK – EO 001.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCT

Arrangements and Procedures

2. EC 001.01 will be a Practical assessment based on content from EO 001.01.
3. Pass/Fail will be based on the EC check list.
4. Time required: 2 x 50 min periods
5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a Practical assessment based on content from EO 001.01.
2. If you have any questions during this assessment, raise your hand.
3. References will be provided.
4. Pass/Fail will be based on the EC check list.
5. You will have 2 x 50 min periods to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
Maintain ordered depth			
Maintain 1 deg bow down angle			

The trainee must achieve 2 SATs to pass this assessment.

EO 001.02 Conduct Dived SOPs as SCOOW

1. Performance. (T0007) Conduct Dived SOPs as SCOOW

2. Conditions

   a. Given:

      (1) Classroom;

      (2) WHDS Trainer; and

      (3) SCT/MCT.

   b. Denied: Nil

   c. Environmental:

      (1) Classroom; and

      (2) SCT.

3. Standard. The trainee shall, IAW references, (T0007) conduct dived Standard Operating Procedures (SOPs) as Ship's Control Officer of the Watch (SCOOW), WRT:

   a. Diving procedures;

   b. Procedures for dived operations;

   c. Miscellaneous routines;

   d. Utilizing Control surfaces in all modes;

   e. Propelling in all modes;

   f. Return to periscope depth;

   g. (T0012) Employ MLD; and

   h. Surfacing procedures.

4. Time Required

   a. EO - 56 x 50 min periods; and

   b. EC - 3 x 50 mins periods.

5. Instructional Methods

   a. 28 x 50 mins Lecture;

   b. 28 x 50 mins Practical; and

c. OJT.

6. Training Aids

a. MLD; and

b. SCANDEX.

7. Test Details. EO 001.02 will be tested in EC 001.02 and PC 001.

8. References. A1, A3, A4, A7, A8, B5, B6, B7, B8, D2, D13, D16, D21, D22, D23, D24

9. Remarks. Two support personnel required for Helmsman and SCC 1 Watchkeeper.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Diving Procedure to include: • Pre Diving Checks to include: ○ OUFD; ○ Diving; ○ Long and short Post Diving Checks; and ○ Reports to the Captain.	Interactive Lecture Practical	3 Periods 7 Periods	D24, D2
TP2	Precautions and Procedures for Sub Dived Ops to include: • Snorting Parameters; • Putting on a Snort; • Breaking the Vacuum; • Reports to the CO and OOW; • Interdependencies of engine loading, battery charging, and SM speed; • Maintain the Snort; and • Stop Snorting Procedure.	Interactive Lecture Practical	3 Periods 7 Periods	D24, D2, A3, B8

TP3	Miscellaneous Routines to include: • Extreme weather precautions • BWA Procedures • Domestic Routines; • Ditching Gash; • Managing Bilge Water; • Maintain Atmosphere; • Pumping Battery Sumps; • Shutting off and Opening up from Attack and Counter Attack; • Conduct an Air Charge; and • Five Man Chamber Procedures.	Interactive Lecture	9 Periods	D24, D2, A4, B6, D13
TP4	SSE Operations to include: • Methods of operation; • Stores; • Caps and Lims; • Operating Procedures; and Bathymetric Requirements	Interactive Lecture Practical	2 Periods 1 Period	A1, A7, D2, D24
TP5	WHDS Operations to include: • Rear Door Procedures; • TOC Operation; • Operating Parameters; and • Water Shot Procedures	Interactive Lecture	3 Periods	D22, D24, D2

TP6	Control Surface Utilization to include: • Planing • Principles • Modes of Operation • Techniques to include: Negative ration planning, considerations for high speed, considerations for slow speed • Rough weather considerations	Interactive Lecture Practical	2 Periods 7 Periods	A8, B5, B7, D2, D24
TP7	Main Propulsion System to include: • Propulsion Mode Selection; • Propulsion Modes; • Propulsion Power Statement; and • Caps and Lims.	Interactive Lecture	2 Periods	B7, D23, D24, B5
TP8	Returning to Periscope Depth Procedures to include: • Preparations; • Equipment Lineups; and • Personnel Requirement.	Interactive Lecture	1 Period	D2, D24, D16, D21
TP9	(T0012) MLD • Principles • Limitations • Implementation	Interactive Lecture Demo	1 Period	A8, D24
TP10	Surfacing Procedure to include: • Surfacing Methods; • Preparations; and • Surfacing.	Interactive Lecture Practical	2 Periods 6 Periods	D2, D24

ENABLING CHECK – EO 001.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.02 is a written test designed to assess the trainee's ability to conduct SOPs as SCOOW. The assessment is based on content from EO 001.02.

3. Pass mark will be 80%.

4. Time required: 3 x 50 min periods.

5. References will not be provided.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.02 is a written test designed to assess your ability to conduct SOP's as SCOOW. The assessment is based on the content from EO 001.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will not be provided.

9. Pass mark is 80%

10. You will have 3 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – ENABLING CHECK LIST

N/A

EO 001.03 Conduct Dived EOPs as SCOOW

1. Performance. (T0006) Conduct Dived EOPs as SCOOW.

2. Conditions

   a. Given:

      (1) Classroom

      (2) SCT/MCT

      (3) References

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) SCT.

3. Standard. The trainee shall, IAW references, (T0006) conduct Emergency Operating Procedures (EOPs) as Ship's Control Officer of the Watch (SCOOW), by:

   a. Conducting Initial SCOOW actions;

   b. Conducting turnover of incident to HQ1;

   c. Taking follow-up EOP action; to include; and

      (1) System restoration; and

      (2) Falling out from Emergency Stations.

4. Time Required

   a. EO - 84 x 50 min periods; and

   b. EC - 16 x 50 min periods (2 per trainee).

5. Instructional Methods

   a. Lecture 14 x 50 min periods;

   b. Practical 70 x 50 min periods; and

   c. OJT.

6. Training Aids. SCANDEX

7. Test Details. EO 001.03 will be tested in EC 001.03 and PC 001.

8. References. A2, A4, D9, D10, D23, D24

9. Remarks. Two support personnel required for Helmsman and SCC 1 Watchkeeper.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Conducting Initial Ship Control Actions to include: • At sea EOPs IAW VCEOPs;	Interactive Lecture/ Practical	14/ 70 periods	A2, D23, D24
TP2	Coordinate Handover Incident to HQ1 to include: • Handover Prereqs; • Duties and Responsibilities ; and • Communications during Emergency Stations.	Interactive Lecture/ Practical	"	D23, D24, A2
TP3	Take Follow-up Actions for EOPs to include: • Coordination of Repair/ Isolation Activities; • Command Briefing Cards; • Compartment Kill Cards; • Engineering Implications; • SCANDEX; • System turnover with HQ1; • Reactivating Systems; • Atmospheric Monitoring; • Electrical Audit; and • Damage Checks.	Interactive Lecture/ Practical	"	D9, D10, D23, D24, A2, A4
TP4	Prereqs for Falling out from Emergency Stations in order to return Submarine to Normal SOP Line up/ Safe State.	Interactive Lecture/ Practical	"	D23, D24
TP5	Smoke Clearance Procedures IAW Smoke Clearance Cards.	Interactive Lecture/ Practical	"	A4, D23, D24

ENABLING CHECK – EO 001.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 001.03 will be a Practical test based on content from EO 001.03.
3. Pass mark will be Pass/Fail based on a checklist.
4. Time required: 2 x 50 min period (per trainee).
5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.03 will be a Practical test based on content from EO 001.03.
2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.
3. Do not open the booklet until you are told to do so.
4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.
5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).
6. Read each question carefully, and all responses fully, before answering the question.
7. If you have any questions during the exam, raise your hand.
8. References will be provided.
9. Pass mark is Pass/Fail
10. You will have 2 x 50 min period to complete this exam (per trainee).
11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Each EOP will require a check list. Each EC will test 2 major and 1 minor EOPs.

PART IV – ENABLING CHECK LIST

Note - * denotes critical factor

Trainee must pass all critical factors and 10 out of 14 non-critical factors.

CHECK LIST	YES	NO	COMMENTS
INITIAL EOP ACTIONS (Multiple – Shall Include a Fire)			
Stop Snorting			
Order Mast State (appropriate masts kept raised/lowered as required)			
*__Make Initial EOP Pipe__			
*__Break Out and Test EBS__			
***Ensure Emergency Crash Stops Made**			
***Ensure Watertight Integrity Correct** (Hatches/Hull Valves as required)			
***Ensure Fire Fighting Systems Available** (Trim System/Fitted System)			
Ensure Bulkheads Shutdown			
Verify state of Bulkhead Doors /Smoke curtains. Take action as necessary			
Line up VCS 446			
***Ensure Appropriate Isolations Made** (GIVs for HP Air Burst, FIVs/Hydraulic Pumps for Hydraulic Bursts)			
***Make Ship Control Orders** ("Blow Main Ballast, Standby to Operate Ship's Systems in Hand, Propel in Manual Mode 1, Zero Hold the Fore planes, After planes in Air Emergency", etc.)			

CHECK LIST	YES	NO	COMMENTS
***Con in Emergency** (EGD, Control Surface Failure, Propulsion Failure)			
Order Damage Checks (if required)			
Ensure Electrical System Implications Considered			
Utilize SCANDEX to ensure EOP actions complete			
Turn over incident to HQ1			
Follow Up EOP Actions			
Brief System Losses			
Ensure System Restorations Made (prompt HQ1 for Hydraulic/Air systems, etc.)			
Conduct Smoke Clearance			
Ensure Systems Reverted to Normal Line-up			
Fall Out from Evolution			

Assessor's Additional Remarks:

Overall Assessment:

YES	NO
PASS	**FAIL**

Trainee Signature

Assessor Signature

This Form to be retained on course files IAW CFNOSSSO'S

PERFORMANCE CHECK – PO 001

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCT

Arrangements and Procedures

2. PC 001 will be a Practical test based on content from PO 001

3. Pass mark will be Pass/Fail

4. Time required: 2 x 50 min period (per trainee)

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

6. PC 001 will be a Practical test based on content from PO 001

7. During this practical assessment, you will be required to Supervise Ship Control. You must pass this assessment to pass PO 001. To pass the test, you must complete the performance statement in accordance with the standard. Do not open the booklet until you are told to do so.

8. References will be provided.

9. Pass mark is Pass/Fail

10. You will have 2 x 50 min period (per trainee) to complete this assessment.

11. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

Note - * denotes critical factor: **TRAINEE MUST PASS ALL CRITICAL FACTORS AND 17 OF 24 NON CRITICAL FACTORS.**

PERFORMANCE FACTORS	YES	NO	COMMENTS
SOP's			
Conduct Watch Turnover			
Conduct Pre-Diving Checks			
Dive Submarine			
Conduct Long Post Diving Checks			
Conduct Depth Change			
Apply MLD			
Apply Propulsion Power Statement			
Conduct Return to Periscope Depth			
Put on a Snort			
Conduct Domestic Routines			
INITIAL EOP ACTIONS (Multiple – Shall Include a Fire)			
Stop Snorting			
Order Mast State (appropriate masts kept raised/lowered as required)			
*__Make Initial EOP Pipe__			
*__Break Out and Test EBS__			
***Ensure Emergency Crash Stops Made**			
***Ensure Watertight Integrity Correct** (Hatches/Hull Valves as required)			
***Ensure Fire Fighting Systems Available** (Trim System/Fitted System)			

PERFORMANCE FACTORS	YES	NO	COMMENTS
Ensure Bulkheads Shutdown			
Verify state of Bulkhead Doors /Smoke curtains. Take action as necessary			
Line up VCS 446			
***Ensure Appropriate Isolations Made** (GIVs for HP Air Burst, FIVs/Hydraulic Pumps for Hydraulic Bursts)			
***Make Ship Control Orders** ("Blow Main Ballast, Standby to Operate Ship's Systems in Hand, Propel in Manual Mode 1, Zero Hold the Fore planes, After planes in Air Emergency", etc.)			
***Con in Emergency** (EGD, Control Surface Failure, Propulsion Failure)			
Order Damage Checks (if required)			
Ensure Electrical System Implications Considered			
Utilize SCANDEX to ensure EOP actions complete			
Turn over incident to HQ1			
Follow Up EOP Actions			
Brief System Losses			
Ensure System Restorations Made (prompt HQ1 for Hydraulic/Air systems, etc.)			
Conduct Smoke Clearance			
Ensure Systems Reverted to Normal Line-up			
Fall Out from Evolution			

Assessor's Additional Remarks:

Overall Assessment:

YES	NO
PASS	**FAIL**

Trainee Signature

Assessor Signature

This Form to be retained on course files IAW CFNOSSSO'S

PO 002 Conduct Dived Operations

EO 002.01 Interpret Submarine Sensor Data

1. Performance. (T0002) Interpret Submarine Sensor Data.

2. Conditions

   a. Given:

      (1) Classroom;

      (2) SCTT; and

      (3) TWS.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall, IAW references, (T0002) interpret Submarine sensor data WRT:

   a. EW systems;

   b. Radar systems; and

   c. Sonar systems.

4. Time Required

   a. EO - 14 x 50 mins periods;

   b. EC - 2 x 50 mins periods; and

   c. PC - 8 x 50 min periods.

5. Instructional Methods

   a. 9 x 50 mins lectures; and

   b. 5 x 50 mins practical.

6. References. A5, B10, B11, B13, B14, B17, D7, D9, D11, D12, D15, D16, D17, D19

7. Training Aids. Nil.

8. Test Details. EO 002.01 will be tested in EC 002.01 and PC 002.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	EW Equipment and theory to include: • Basic Theory; • Capabilities and Limitations; • Frequency; • Mast States; • Danger Levels; and • Operator Drills.	Lecture Practical	2 Periods 1 Period	D19, A5, D9, D17
TP2	Radar systems to include: • Capabilities and Limitations; • Automatic and Manual Tracking; • Operator Drills; • Operator Reports; and • SDM Interface.	Lectures Practical	2 Periods 1 Period	D12, D15, D11
TP3	Sonar systems to include: • Capabilities and Limitations; • Sonar Fit; • Active Operations; • Passive Operations; • Intercept Operations; • Automatic and Manual Tracking; • Operator Drills; and • WSDB Interface.	Lectures Practical	4 Periods 3 Periods	D16, B10, B11, B13, B14, B17
TP4	Automatic Information System (AIS) to include: • Capabilities and Limitations • Operation	Lectures	1 Period	D7

ENABLING CHECK – EO 002.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.01 will be a Written test based on content from EO 002.01.

3. Pass mark will be 80%.

4. Time required: 2 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a written test based on content from EO 002.01.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%.

10. You will have 2 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

N/A

EO 002.02 Conduct Broadband Target Motion Analysis

1. Performance. Conduct Broadband Target Motion Analysis (TMA).

2. Conditions

   a. Given:

      (1) Classroom;

      (2) VSCTT; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall, IAW references, conduct Broadband TMA by:

   a. Interpreting Bearing Curves;

   b. Determining Contact Solutions;

   c. Calculating Single Leg Ekelund Ranges;

   d. Calculating Double Leg Ekelund Ranges;

   e. Calculating Pboot;

   f. Calculating Min/ Max Ranges;

   g. Utilizing LOP Bearing Only Plotting Techniques;

   h. Plotting Visual contacts; and

   i. (T0003) Employing SFCS.

4. Time Required

   a. EO - 39 x 50 min periods; and

   b. EC - 2 x 50 mins periods.

5. Instructional Methods

   a. 22 x 50 mins periods lecture; and

   b. 17 x 50 mins periods practical.

6. Training Aids

   a. Bearing Rate Computer;

b. SDM Laptops (1 per trainee);

c. Plotting Paper; and

d. Scientific calculator.

7. Test Details. EO 002.02 will be tested in EC 002.02 and PC 002.

8. References. A6, B19, D1, D15, D20.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Bearing Curve interpretations: • Classic; • Basic; • Target Zigs; and • CPA solution.	Interactive Lecture	1 Period	D15

TP2	Determining Contact Solutions
	• RBROTSARC tote;
	• Line of Sound Diagram:
	  o OSA;
	  o TSA; and
	  o RSA.
		Interactive Lecture	4 Periods
			Home Assignment
				D15

TP3	Single Leg Ekelund Ranging: • Procedure; • Calculation; and • Block report.	Interactive Lecture	2 Periods	D15

TP4	Double Leg Ekelund Ranging:
	• Principles;
	• Considerations:
	  ○ OSA change; and
	  ○ Manoeuvring.
	• Procedure;
	• Calculation;
	• Ballpark range; and
	• Block report.
		Interactive Lecture
		Practical
			Home Assignment
			7 Periods Lecture
			3 Periods Practical
				D15

TP5	Pboot Calculation: • Principles; • Procedure; • Calculation; • Block report; and • Pboot for a given range.	Interactive Lecture Practical	1 Period Lecture 2 Periods Practical Home Assignment	D15
TP6	Min/ Max Contact Ranges • Manipulating SLE formula	Interactive Lecture	1 Period	D15

TP7	LOP Bearing Only Plotting Techniques:
	• Principles;
	• Bearing Fan Interpretation;
	• Procedures:
	  ○ Plotting Contacts;
	  ○ Zig procedures; and
	  ○ CQD
	• Calculations;
	• Block Report;
	• Min/ Max Range for given speeds; and
	• Crossover Ranges.
		Interactive Lecture
		Demonstration Practical
			3 Periods Lecture
			4 Periods Practical
				D15, D20

TP8	Plotting Visual Contacts: • Target Setup; • Speed between the looks; • Completing the SAMO tote; and • Block Report.	Interactive Lecture Demonstration Practice	1 Period 3 Periods	D1, D15
TP9	CCS 876 • Architecture; • Information Flow; • Capabilities and Limitations; • Operation and Functions; • Page Management; • RMS; • PBOOT; • Zig Procedure; • Ranging Manoeuvres; • Visual Setups • Track Management; and • Snap Firing.	Interactive Lecture Practical	2 Periods Lecture 5 Periods Practical	D20, A6, B19

ENABLING CHECK – EO 002.02

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.02 will be a written test based on content from EO 002.02.

3. Pass mark will be 80%.

4. Time required: 2 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.02 will be a written test based on content from EO 002.02.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%.

10. You will have 2 x 50 min period to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

N/A

EO 002.03 Conduct Narrowband Target Motion Analysis

1. Performance. Conduct Narrowband (NB) Target Motion Analysis (TMA)

2. Conditions:

   a. Given:

      (1) Classroom;

      (2) VSCTT; and

      (3) References.

   b. Denied: Nil.

   c. Environmental: Classroom.

3. Standard. The trainee shall, IAW references, conduct Narrowband TMA by:

   a. Resolving bearing ambiguity;

   b. Interpreting frequency relationships;

   c. Calculating a reasonable base frequency;

   d. Calculating cone of courses;

   e. Calculating frequency rate Ekelund range;

   f. Determining contact solution (range +/- 20%, course +/- 20 degrees, speed +/- 3 knots); and

   g. Announcing contact solution.

4. Time Required

   a. EO - 38 x 50 min periods; and

   b. EC - 2 X 50 min periods.

5. Instructional Methods

   a. 18 x 50 mins Periods Lecture; and

   b. 20 x 50 mins Periods Practical.

6. Training Aids

   a. 1959 Range Rule; and

   b. Scientific calculator.

7. Test Details. EO 002.03 will be tested in EC 002.03 and PC 002.

8. References. A6, B17, D6, D15, D16, D20.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Characteristics affecting the Towed Array (TA): • Beam shape/ forming; • Beam Width; • Array stabilization time; • Array cant; • Array depth; • Depression/ Elevation ambiguity; and • Parallax Error	Interactive Lecture	2 periods	D6, D16, B17
TP2	Resolving Bearing Ambiguity: • Altering for NB contacts; and • Observed effects	Interactive Lecture	1 period	D6, D15, D16
TP3	Triangulation BB Ranging	Interactive Lecture	1 period	D6, D15, D16
TP4	Narrowband Definitions: • Frequency Terms	Interactive Lecture	2 periods	D6, D15, D16
TP5	Doppler Effect	Interactive Lecture	1 period	D6, D15, D16
TP6	LOS Diagram	Interactive Lecture	1 period	D6, D15, D16
TP7	Frequency Relationships: • Fc vs Fe; • Fr vs Fc; • Fr vs Fe; • Fr = Fe; • Fe = Fc; and • Cones of courses.	Interactive Lecture	3 periods	D6, D15, D16

TP 8	Frequency Calculations • Calculation of Fo, Fr, Fc; and • PFoot.	Interactive Lecture / Demo Performance	2 periods 7 periods	D6, D15, D16
TP 9	Range Calculations: • Frequency Rate Ekelund; • Rate of Change of Range Rate; and • Range under the curve.	Interactive Lecture / Demo Performance	3 periods 7 periods	D6, D15, D16
TP10	BFP Plotting Techniques: • Setup; • Scales; • Symbology; and • 1959 Range Rule.	Interactive Lecture / Demo Performance	1 period 3 periods	D6, D15, D16
TP11	CCS 876 Towed Array Function	Demo Performance/ Simulation	1 period 3 periods	D6, D15, D16, D20, A6

ENABLING CHECK – EO 002.03

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.03 will be a written test based on content from EO 002.03.

3. Pass mark will be 80%.

4. Time required: 2 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.03 will be a written test based on content from EO 002.03.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%.

10. You will have 2 x 50 min periods to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

N/A

EO 002.04 Employ External Communication Systems

1. Performance. (T0001) Employ external communication systems

2. Conditions

   a. Given:

      (1) Classroom;

      (2) Communications Training Facility (CTF); and

      (3) VSCTT.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) VSCTT.

3. Standard. The trainee shall, IAW references, (T0001) employ external communications systems WRT:

   a. RF communications;

   b. Submarine Broadcast procedures;

   c. Underwater Telephone (UWT);

   d. Voice procedures; and

   e. (T0013) Breaking down operational messages.

4. Time Required

   a. EO - 17 x 50 min periods; and

   b. EC - 2 x 50 min periods EC.

5. Instructional Methods

   a. 10 x 50 mins periods lectures; and

   b. 7 x 50 mins periods practical.

6. Training Aids. Classified IT with DND Message Drafter software.

7. Test Details. EO 002.04 will be tested in EC 002.04 and PC 002.

8. References. B1, B2, B3, B4, B10, B18, D2, D4, D5, D8, D16.

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	RF Communications • Fitted Equipment; • Capabilities and Limitations; • Frequency Bands; • System Line-ups; • Antenna Line-ups; • Cryptographic Equipment; • NUCO/Authentication; and • ECBs.	Interactive Lecture	2 Periods	D4, D8, B18
TP2	Submarine Broadcast Procedures: • Routines; • Broadcast Control Authority; • Operator Reports; and • Submarine Accountability Messages.	Interactive Lecture	2 Periods	D4, D5
TP3	UWT: • Equipment; • Capabilities and Limitations; • UWT Codes; • Ranging Methods; • Uncle Joe Procedure; • Hooker Procedure; and • Operator Reports.	Interactive Lecture Practical	1 Period 1 Period	D16, B10, B4
TP4	Voice Procedures: • Brevity Codewords; • Secure/Non-secure Lines; and • Remote Control Unit.	Interactive Lecture Practical	1 Period 3 Period	B2

TP5	(T0013) Breakdown of Operational Messages:
	• SUBNOTE to include:
	  ○ SUBNOTE Req;
	  ○ SUBNOTE chg; and
	  ○ SUBNOTE Chg Req.
	• Diving Message;
	• RSS/ESR;
	• CASEX to include:
	  ○ Starred relaxation req.
	• Daily SITREP;
	• MARINTREP;
	• BARNSTORM;
	• OPSTAT Unit; and
	• RFI/RFS.
		Interactive Lecture
		Practical
			4 Periods
			3 Periods
				B1, B4, D5, D2, B3

ENABLING CHECK – EO 002.04

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.04 will be a written test based on content from EO 002.04.

3. Pass mark will be 80%.

4. Time required: 2 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.04 will be a written test based on content from EO 002.04.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%.

10. You will have 2 x 50 min periods to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

N/A

EO 002.05 Coordinate the Watch

1. Performance. (T0010) Coordinate the Watch.

2. Conditions

   a. Given:

      (1) Classroom;

      (2) References; and

      (3) SCTT;

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) VSCTT.

3. Standard. The trainee shall, IAW references, (T0010) coordinate the watch by:

   a. Conducting watch turnover;

   b. Employing signature management;

   c. Maintaining records;

   d. Applying environmental regulations; and

   e. Employing CIO.

4. Time Required

   a. EO - 12 x 50 min periods; and

   b. EC - 2 x 50 min periods.

5. Instructional Methods

   a. 10 x 50 mins periods lecture;

   b. 2 x 50 mins periods practical; and

   c. OJT.

6. Training Aids. Nil.

7. Test Details. EO 002.05 will be tested in EC 002.05 and PC 002.

8. References. B21, D1, D2, D5, D6, D13, D14, D15, D21, D22, D24, D25

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Watch Turnover: • Pre-watch Rounds; • Post-watch Rounds; • Captain's Night Orders; • Navigation; • Sensors; • Contact Situation; • Ship Control; • Equipment Considerations; • Watch Brief; and • Daily User Checks.	Interactive Lecture	2 Periods	D24, D2, D22, D25
TP2	Employing Signature Management to include: • Organization; • Purpose; • Quiet States; and • Noise Husbandry.	Interactive Lecture	1 Period	D24, D25, D2, D6
TP3	Maintaining Records to include: • Operational Records; and • Logs.	Interactive Lecture	2 Periods	D15, D2, D24
TP4	Applying Environmental Regulations.	Interactive Lecture	1 Period	D13, D2, D24, D14
TP5	Employing CIO to include: • Personnel Required; • Duties and Responsibilities; and • Information Flows.	Interactive Lecture	2 Periods	D15, D1, D2, D25
TP6	Close Quarters Drill (CQD)	Interactive Lecture Practical	1 Period 1 Period	D15, D2, D21

TP7	Fishing Vessel Avoidance	Interactive Lecture Practical	1 Period 1 Period	B21, D5, D15, D2

ENABLING CHECK – EO 002.05

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.05 will be a written test based on content from EO 002.05.

3. Pass mark will be 80%.

4. Time required: 2 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.05 will be a written test based on content from EO 002.05.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%.

10. You will have 2 x 50 min periods to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

N/A

EO 002.06 Calculate Submarine Position

1. Performance. (T0005) Calculate Submarine Position.

2. Conditions

   a. Given:

      (1) Classroom; and

      (2) SCTT.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) VSCTT.

3. Standard. The trainee shall, IAW references, (T0005) calculate Submarine position utilizing:

   a. Electronic navigation aids;

   b. WSM/PMI techniques; and

   c. Echo Sounders.

4. Time Required

   a. EO - 32 x 50 min periods; and

   b. EC - 3 x 50 min periods.

5. Instructional Methods

   a. 10 x 50 min periods Lectures; and

   b. 22 x 50 min periods Practical.

6. Training Aids. SDM Laptops (8).

7. Test Details. EO 002.06 will be tested in EC 002.06 and PC 002.

8. References. B4, B9, B12, B16, D3, D5, D7, D11, D16, D20.

9. Remarks. Dependant on availability, book tour of Navigation Trainer for KH-1007 familiarization.

10. Teaching Points

	Description	Method	Time	Reference

TP1	SDM • Architecture; • Operation; • Caps and Lims; and • Speed-Times-Distance Functions.	Interactive Lecture Practical	4 Periods 12 Periods	D11, D7, B12
TP2	Display and interpret Operational Information to include: • SOA; • WSM; and • PMI	Interactive Lecture	1 Period	D20, B4, D5

TP3	Calculate Submarine position to include:
	• Snap Sounding;
	• Bottom Contour Fixing to include:
	  ○ Safe depth; and
	  ○ Safety separation zone.
	• Passage Graphs;
	• POE to include:
	  ○ Principles;
	  ○ Electronic POE Theory; and
	  ○ Reduction methods/ Limitations
		Interactive Lecture
		Practical
			4 Periods
			10 Periods
				B9, B16, D16, D3, D5, B12

TP4	Echo Sounders • Equipment • Capabilities and Limitations	Interactive Lecture	1 Period	B9, B16, D16

ENABLING CHECK – EO 002.06

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. EC 002.06 will be a written test based on content from EO 002.06.

3. Pass mark will be 80%.

4. Time required: 3 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.06 will be a written test based on content from EO 002.06.

2. Ensure your desk is clear of material except for the test booklet and/or equipment necessary for this exam.

3. Do not open the booklet until you are told to do so.

4. Write your name on the answer sheet and all scrap paper. Ensure all scrap paper is attached to the appropriate answer sheet.

5. Work carefully. Do not spend too much time trying to answer question(s) with which you have difficulty. Continue the exam then come back to the problem question(s).

6. Read each question carefully, and all responses fully, before answering the question.

7. If you have any questions during the exam, raise your hand.

8. References will be provided.

9. Pass mark is 80%.

10. You will have 3 x 50 min periods to complete this exam.

11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

N/A

PERFORMANCE CHECK – PO 002

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. VSCTT

Arrangements and Procedures

2. PC 002 will be a practical test based on content from PO 002.

3. Pass mark will be Pass/ Fail based on a checklist.

4. Time required: 8 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical test based on content from PO 002.

2. During this practical assessment, you will be required to Conduct Dived Operations. You must pass this test to pass PO 001. To pass the test, you must Conduct Dived Operations in accordance with the standard.

3. References will be provided.

4. Pass mark is “Pass”, and is based on a checklist.

5. You will have 8 x 50 min periods to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

Note - * denotes critical task

TRAINEES MUST PASS ALL CRITICAL TASKS AND 18 OF 25 NON-CRITICAL TASKS

CHECK LIST	YES	NO	COMMENTS
SDM			
***Input Pool of Errors into SDM using given parameters**			
Reduce POE			
***Plot assigned areas into SDM**			
***Ensure the submarine remains within assigned waters**			
SFCS Drills			
Complete closing up checks on SFCS			
Conduct RMS			
Conduct PBOOT			
Conduct Ranging Manoeuvre			
Conduct ZIG Procedures			
Input Visual Set Up			
Conduct Snap Firing			
Conduct Track Management (Nominate threats and Designate tracks)			
Manual TMA			
***Assess opening or closing curves from TBP**			
Calculate DOT			

CHECK LIST	YES	NO	COMMENTS
Calculate Range by Single Leg Eklund			

CHECK LIST	YES	NO	COMMENTS
***Calculate Range by Double Leg Eklund**			
Calculate Range at CPA			
Perform as LOPO			
Set up the LOP			
Use appropriate range scale			
Use proper plotting conventions			
Conduct bearing only analysis on 1 Contact			
Determine maximum/minimum ranges for given speeds			
Plot target setup			
Conduct Zig Procedure			

Assessor's Additional Remarks:

Overall Assessment:

YES	NO
PASS	**FAIL**

Trainee Signature

Assessor Signature

This Form to be retained on course files IAW CFNOSSSO'S

PO 003 Conduct Periscope Watchkeeping

EO 003.01 Maintain Submarine Safety at Periscope Depth

1. Performance. Maintain Submarine Safety at Periscope Depth

2. Conditions

   a. Given:

      (1) VSCTT;

      (2) Classroom; and

      (3) Stop Watch.

   b. Denied: Nil.

   c. Environmental:

      (1) Classroom; and

      (2) VSCTT.

3. Standard. The trainee shall, IAW references, maintain Submarine safety at periscope depth, by:

   a. Operating the periscope;

   b. Adhering to periscope safety procedures; and

   c. Conducting periscope drills

   d. Conducting periscope watch activities.

4. Time Required

   a. EO - 72 x 50 min periods;

   b. EC - 4 x 50 min periods; and

   c. PC - 7 x 50 min periods.

5. Instructional Methods

   a. 11 x 50 mins Periods Lecture; and

   b. 61 x 50 mins Periods Practical.

6. Training Aids. Stop Watch.

7. Test Details. EO 003.01 will be tested in EC 003.01 and in PC 003.

8. References. B3, B15, B20, D1, D2, D12, D15, D18, D19, D21, D23

9. Remarks. Nil.

10. Teaching Points

	Description	Method	Time	Reference
TP1	Periscope Watch Keeping: • Turnover; • All round Look Techniques; • Capabilities and Limitations; • Mast States; • Mast Exposure Limits; • Raising and Lowering Speeds; • Sinuation; • Periscope Imagery; • Operation • Ranging Techniques • ATB Estimation • Target Setup Drill/ Range For Me • Shipping Reports to OOW/CO	Interactive Lectures	7 Periods	D1, D15, B15, D2, D18, B20, B3
TP2	Periscope Safety: • Emergency Go Deep; • Go Deep Circle; and • Range Rates.	Interactive Lectures	2 Periods	D23, D2, D15
TP3	Watch Activities: • Warner Clearance; • Radar Cut Procedure; • Warner Look; • Warner Take; and • Close Quarters Contact.	Interactive Lectures Practical	2 Periods 61 Periods (for all TPs)	D21, D2, D12, B3, D19

ENABLING CHECK – EO 003.01

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. VSCTT

Arrangements and Procedures

2. EC 003.01 will be a practical test based on content from EO 003.01.

3. Pass mark will be 100%, and is based on a checklist.

4. Time required: 4 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. EC 003.01 will be a Practical test based on content from EO 003.01.

2. During this practical assessment, you will be required to Conduct Periscope Watchkeeping. You must pass this test to pass EO 001.01. To pass the test, you must Conduct Periscope Watchkeeping in accordance with the standard.

3. References will be provided.

4. Pass mark is 100%, and is based on a checklist.

5. You will have 4 x 50 min periods to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Nil

PART IV – ENABLING CHECK LIST

PART IV – ENABLING CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
1. Conduct a periscope turnover.			
2. Keep the ARL (Day, Night and Restricted Visibility).			
3. Initiate an Emergency Go Deep.			
4. Range a contact +/- 1 Kyd.			
5. Make a shipping report to OOW.			
6. Conduct a visual set up.			
7. Conduct Warner Clearance.			
9. Conduct Radar Cut Procedure.			

Note: Trainee must achieve SAT for all items to pass.

PERFORMANCE CHECK – PO 003

PART 1 – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. SCTT

Arrangements and Procedures

2. PC 003 will be a practical test based on content from PO 003

3. Pass mark will be Pass/ Fail based on a checklist.

4. Time required: 7 x 50 min periods.

5. References will be required.

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003 will be a practical test based on content from PO 003

2. During this practical assessment, you will be required to conduct periscope watchkeeping. You must pass this test to pass PO 001. To pass the test, you must conduct periscope watchkeeping in accordance with the standard.

3. References will be provided.

4. Pass mark is “Pass”, and is based on a checklist.

5. You will have 7 x 50 min periods to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Nil

PART IV – PERFORMANCE CHECK LIST

CHECK LIST	SAT	UNSAT	COMMENTS
1. Conduct a periscope turnover.			
2. Keep the ARL (Day, Night and Restricted Visibility).			
3. Initiate an Emergency Go Deep.			
4. Range a contact +/- 1 Kyd.			
5. Make a shipping report to OOW.			
6. Conduct a visual set up.			
7. Conduct Warner Clearance.			
9. Conduct Radar Cut Procedure.			

Note: Trainee must achieve SAT for all items to pass.

Assessor's Additional Remarks:

Overall Assessment:

YES	NO
PASS	**FAIL**

Trainee Signature

Assessor Signature

This Form to be retained on course files IAW CFNOSSSO'S

ANNEX A – MAIN REFERENCES

A1 = CF or other official Canadian government publication

B = Official Allied Force or Allied government publication

C = Commercial publication

D = Other

Ref Code	Publication
A1	C-09-153-003/TS-000 Explosives Safety Manual Vol 3 - Ships
A2	C23-045-000/AG-004 Damage Control (Sea) Vol 4 – Submarine Damage Control
A3	C-26-509-000/MS-001 Technical Operations Manual Submarine Main Battery type 37KR177T
A4	C-29-537-000/MS-001 Air Purification in Submarines (BR 1326)
A5	C-58-407-000/MS-004 Sea Search
A6	C-69-815-000/MS-001 SFCS
A7	C-74-370-E00/TA-000 Description and Maintenance Instruction Submarine Launched Pyrotechnics
A8	B-GN-040-000/TS-002-4 MLD Manoeuvring Limitation Diagram
B1	App II Allied Maritime Voice Reporting Procedures / Structured Messages
B2	APP 7 Joint Brevity Code Words Publication
B3	ATP 18 Allied Manual of Submarine Operations
B4	Allied ECM Vol II AXP 1 Allied Submarine and Anti Submarine Exercise Manual
B5	BR 1966 (21) 002 One Man Control Console
B6	BR 1966 (27) Buoyant Wire Aerial
B7	BR 1966 (9) 02 Introduction to Power Generation, Propulsion and Control
B8	BR 2075 K Main battery Chloride 880T
B9	BR2800 Echo Sounder Type 778

B10	BR 2805 Sonar Type 2008
B11	BR 2805 Sonar type 2040
B12	BR 45 (4) Admiralty Manual of Navigation
B13	BR 8151 Sonar Type 2040
B14	PRS Operator Manual
B15	BR 8409 Periscope System UPHOLDER Class
B16	BR 8551 (8) Echo Sounder Type 780
B17	BR 9072 Sonar Type 2046
B18	Comms System Block Diagram DMSS 8-3 Sketch No. 7003B BR 222 (SM)
B19	FMS1 MK 48 Reference Manual Vol1
B20	SMP 68 Conduct of Diesel Submarine Operations
B21	SMP 95 Fishing Vessel Avoidance
D1	Canadian Submarine Attack Drills
D2	Captain's Standing Orders (CSOs)
D3	CFCD 106 Vol I, II, III MARTIS
D4	CFCD 107 Naval Communications
D5	CFCD 110 Submarine Operating instructions
D6	CFCD 113 Acoustic Information Guide
D7	CFCD 130 Canadian navigation Manual
D8	CFCD 501 CF Communication Operating Instructions
D9	Command Briefing Cards
D10	Compartment Kill Cards
D11	ECPINS Operator Manual
D12	Kelvin Hughes Radar Type 1007 User Manual Vol I and II
D13	MARCORD G-18 Environmental Management of Shipboard Waste
D14	MARLANT VICTORIA Class EMS

D15	NCPM 401 VICTORIA Class Combat Information Organization
D16	NCPM 403 Canadian Submarine SONAR Operating manual
D17	NOSTP 420 Submarine Electronic Warfare Guide
D18	NOSTP 440 Submarine Periscope Photography Handbook
D19	Sea Search II Operator manual
D20	SFCS Operator Manual
D21	Submarine Tactical Operating Procedures (STOPs)
D22	SUBWOPs
D23	VICTORIA Class Emergency Operating Procedures (EOPs)
D24	VICTORIA Class Standard Operating Procedures (SOPs)
D25	VICTORIA Class Submarine Standing Orders (VCSSOs)

ANNEX B – TASK LIST

TASK NO.	TASK ELEMENT	TRG / NO TRG	PO
T0001	Use External Communications Systems	T	002
T0002	Interpret Submarine Sensor Data	T	002
T0003	Manage Submarine Fire Control System	T	002
T0004	Conduct Periscope Watchkeeping	T	003
T0005	Calculate Submarine's Position	T	002
T0006	Conduct dived Emergency Operating Procedures (EOPs) as Ship Control Officer of the Watch (SCOOW)	T	001, 003
T0007	Conduct dived Standard Operating Procedures (SOPs) as Ship Control Officer of the Watch (SCOOW)	T	001, 002
T0008	Maintain Trim	T	001
T0009	Supervise Ship Control	T	001
T0010	Coordinate the Watch	T	002
T0011	Conduct Dived Operations	T	002
T0012	Employ Manoeuvring Limitation Diagram (MLD)	T	001
T0013	Breakdown operational messages	T	002

ANNEX C – QUALIFICATION STANDARD AND PLAN WRITING BOARD RECORD OF PROCEEDINGS

Dates: 3-14 Dec 12

Location: CFNOS Halifax NS

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Bryan	P	LCdr	00207-02	CFNOS Sub Div
Collins	P	LCdr	00207-02	CFNOS Sub Div
Mudge	K	LT(N)	00363	CFNOS Stds Div
Pischedda	J	LT(N)	00207-02	CFNOS Sub Div
McGuire		CPO2	00114	ST (A)
Girard	M	PO1	00324	CFNOS Stds Div
Janssen	J	PO1	00299	DMTE

Board Objective: The aim of the board was to combine the current COTW (AJSL) SS, SOQC (AILV) SS and SWD (AILT) SS into two new Specialty Specifications (Second Officer of the Watch and Submarine Warfare Director).

Specification Task – No Train Decisions:

The tasks selected for training are detailed at Annex B. Tasks not selected for training are indicated on the accompanying SPQR amendment and creation paperwork.

Discussions / Decisions:

3 Dec 2012

The Chair introduced the board and identified that the aim of the board was to combine the current COTW (AJSL) OSS, SOQC (AILV) OSS and SWD (AILT) OSS into two new OSSs (Second Officer of the Watch and Submarine Warfare Director) and move the navigation tasks to the Surface FNO course as a Submarine PO. Since several members of the board had limited experience with QSP Review Boards; the TDO gave a brief WRT the board process and CFITES role.

The Board started with Task Analysis. They created a scalar using all remaining tasks and reviewed the task lists from the applicable OSSs. During this review, several new tasks were added and/or amended. IAW the new MESIP guidelines all levels and Knowledge statements will be removed. Key points/tasks that were introduced were:

Task statements for the new 2OOW SPQR are as follows:

1) T0001 “Use external communications systems”.

2) T0002 “Interpret Submarine sensor data”.

3) T0003 “Manage Submarine Fire Control System”.

4) T0004 “Conduct periscope watchkeeping”.

5) T0005 “Calculate submarine position”.

6) T0006 “Conduct dived Emergency Operating Procedures (EOPs) as Ship Control Officer of the Watch (SCOOW)”.

7) T0007 “Conduct dived Standard Operating Procedures (SOPs) as Ship Control Officer of the Watch (SCOOW)”.

8) T0008 “Maintain Trim”.

9) T0009 “Supervise Ship Control”.

10) T0010 “Coordinate the Watch”.

11) T0011 “Conduct dived operations”.

12) T0012 “Employ Manoeuvring Limitation Diagram (MLD)”.

13) T0013 “Breakdown operational messages”.

Following a group discussion WRT specific SSs, the following changes were proposed:

The Submarine Warfare Director Qualification (AILT) the following tasks were proposed for deletion:

1) T0001 “Conduct Submarine Navigation

2) T0002 “Conduct Navigation related administration”

3) T0003 “Provide advice to command on navigation related matters”

4) T0004 “Provide advice to command warfare related matters”

5) T0006 “Provide advice to command on the use of sensors”

6) T0007 “Assist in the planning of operations and exercises”

7) T0008 “Assist in the execution of operations and exercises”

8) T0009 “Maintain operational readiness of assigned section”

The Submarine Chief of the Watch Qualification (AJSL) the following tasks were proposed for deletion:

1) T0002 “Apply principles of the broadcast”
2) T0003 “Apply principles of the Antenna line ups”
3) T0004 “Communicate using low grade codes”
4) T0005 “Communicate using voice procedures”
5) T0006 “Communicate using the Under Water Telephone (UWT)”
6) T0011 “Conduct Target Motion Analysis (TMA)
7) T0012 “Utilize data link information
8) T0013 “Utilize Global Command Control System- Maritime (GCCS-M) information”
9) T0015 “Prepare Weapon for firing”
10) T0016 “Fire the weapon”
11) T0017 “Control weapon post firing”
12) T0019 “Recognize Periscope threats”
13) T0020 “Conduct Ranging Techniques
14) T0021 “Conduct Target setup (TSU)”
15) T0022 “Conduct Radar Cut Procedures”
16) T0023 “Conduct Warner Look/Take”
17) T0024 “Conduct Look Procedures”
18) T0025 “Calculate All Around Look (ARL)”
19) T0026 “Calculate Go Deep Ranges”
20) T0027 “Conduct Periscope Photography”
21) T0028 “Use Thermal Imaging (TI)”
22) T0029 “Conduct Emergency Go Deep (EGD)”
23) T0030 “Conduct dived navigation”
24) T0032 “Conduct Bottom Contour Fixing

25) T0033 “Create Pool of Errors (POE)”

26) T0035 “Conduct Snap Soundings”

27) T0041 “Advise Command on Engineering implications”

28) T0043 “Supervise helm”

29) T0045 “Direct daily user checks”

30) T0047 “Provide recommendations to command on submarine teactics”

31) T0050 “Interpret Advanced Refractive Effects Prediction System (AREPS)”

32) T0051 “conduct Pre Watch Control Room rounds”

The Submarine Officer Qualification (AILV) the following tasks were proposed for deletion:

1) T0001 “Conduct Contact Evaluation with Submarine Fire Control System (SFCS)

2) T0002 “Maintain trim”

3) T0003 “Operate Echo Sounder”

4) T0004 “Communicate utilizing external comms”

5) T0005 “Conduct Emergency Operating Procedures (EOPs) as Ship Control Officer of the Watch (SOOCW)”

6) T0006 “Conduct Manual Target Motion Analysis (TMA)”

7) T0007 “Conduct dived navigation”

8) T0008 “Conduct periscope watchkeeping”

9) T0009 “Stand Officer of the Day submarine”

10) T0010 “Direct ship control”

11) T0011 “Interpret Ship’s sensor data”

12) T0012 “Conduct Contact Evaluation”

13) T0013 “Conduct Pre/Post Watch Rounds”

14) T0014 “Evaluate Engineering implications”

15) T0015 “Employ Operational references”

16) T0016 “Direct Duty Watch”

17) T0017 “Direct Duty Watch routines”

18) T0018 “Conduct Duty Watch administration”

19) T0019 “Conduct Submarine Rounds Alongside”

20) T0020 “Coordinate Alongside SOPs”

21) T0021 “Initiate Response to Bomb Threat”

22) T0022 “Initiate Response to Environmental Incident”

23) T0023 “Coordinate Alongside SOPs”

24) T0024 “Direct At Sea Watch Routines”

25) T0025 “Conduct At Sea Watch Administration”

26) T0026 “Conduct Record Keeping”

27) T0027 “Stand as a Control Room Watch Keeping Officer at Sea”

28) T0028 “Navigate with Electronic Navigational Aids”

29) T0029 “Conduct Contact Evaluation with Electronic Navigation Aids”

30) T0030 “Conduct Duty Watch EOP Response”

31) T0031 “Conduct SOPs as SCOOW”

The board agreed on the task statements for the 2 new SSs. DMTE 2-2-2-4 PO1 Janssen approved these changes in principle and the board continued with development as if they were approved. New SPQR paperwork will be submitted in conjunction with the QSPs. Scalar development followed with PO 001 for the SCOOW.

4 Dec 12

Scalar development continued with the 2OOW, and 3 POs were completed. EO 001.01 was started.

5 Dec 12

The board continued with the development of EOs, completed EO 001.01, 02, 03 and PC 001. References will be added at the end of the board.

6 Dec 12

Course development continued with PO 002 and 003 and all associated EOs, ECs and PCs.

7 Dec 12 – The board did not sit this day as there was a CFNOS parade and the Jr Ranks Christmas Dinner.

10 Dec12

The QSPWB shifted to the SMWD QSP and developed PO 001 with its associated EOs, ECs and PC. Some concurrent activity was started to develop a Main Reference List for the 2OOW QSP. PO 002 – PO 005 with their associated EOs, ECs and PCs were developed.

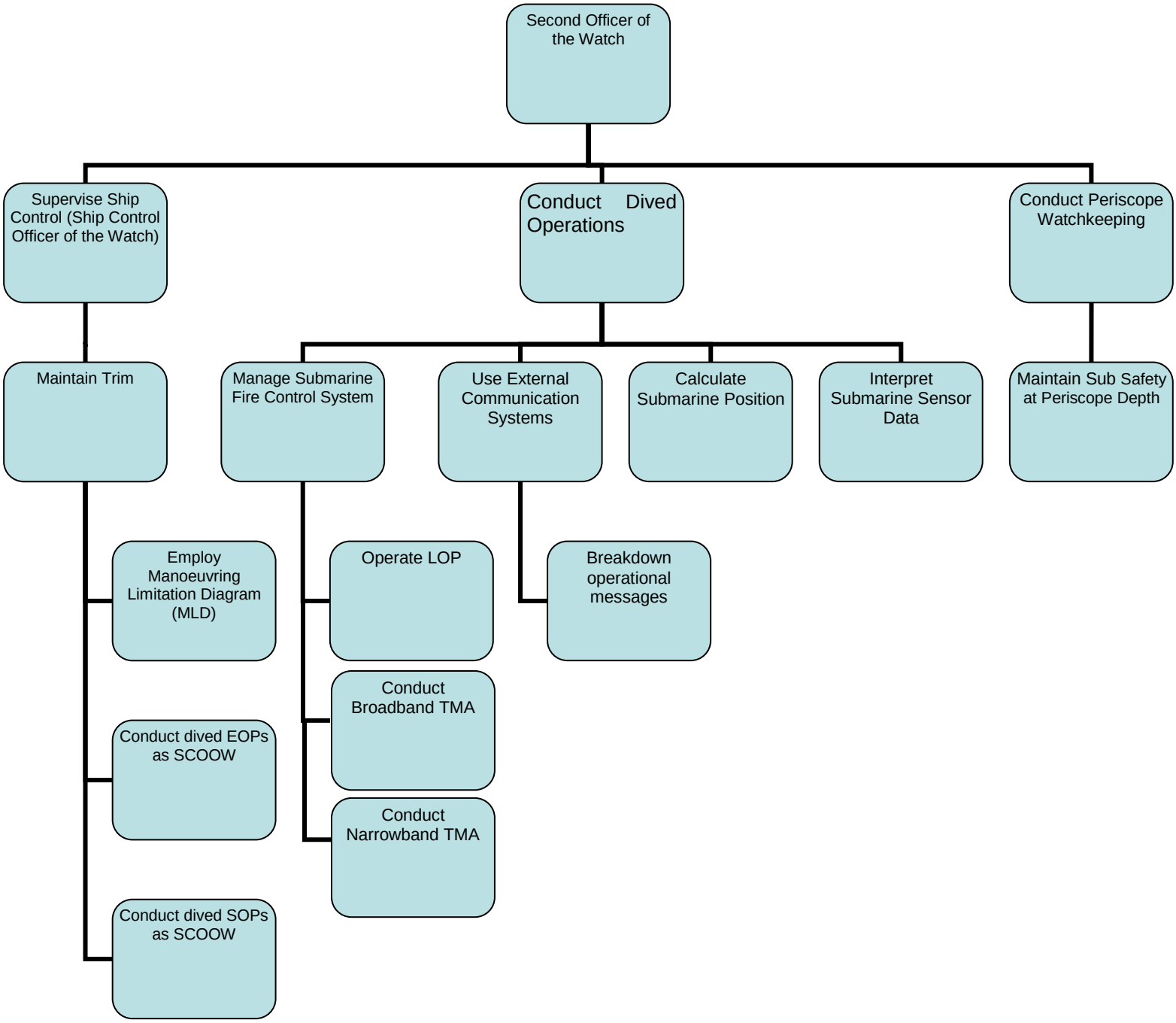
11 Dec 12

Scalars were completed and work continued on the Main Reference List. SPQR paperwork was initiated.

12 Dec 12

Both QSPs were reviewed, SPQR paperwork was reviewed. The main QSPWB was dismissed. Post board activity will continue secretarially.

ANNEX D – QSPWB SCALAR



Approved 04 Apr 2014
Version No 1.0
MITE Crse ID 120231/120232

ANNEX E – ON-JOB PERFORMANCE RECORD

(CERTIFICATE OF QUALIFICATION)

NAME and INITIALS RANK SN

Date Training Started Expected Date of Completion

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The CCA for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance Objective (PO) to the required standard, the Training Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise available. The Training Officer shall insert an “E” (Exception) in the remarks column of the OJPR and include a brief explanation opposite that PO.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under “Additional Duties” at the end of the OJPR.

5. Upon completion of the OJPR sheet (trainee has completed all required POs), the following action will be taken:

a. Fleet Exam (if applicable);

b. Nominate the trainee (by message) for fleet examination:

(1) Forward a completed copy of the Annex C (section 7 completed) to CFNOS Standards Section (as appropriate) at least five (5) days prior to examination date; and

(2) Upon successful completion of the Fleet Exam, the Standards Section is to forward the trainee results by message to the parent unit and DPCOR.

c. Administrative (all candidates):

(1) The Divisional Officer (NCMs) OR School Commandant OR Commanding Officer (Officers) shall complete section 6 of the OJPR Certificate of Qualification;

(2) The individual's Qualification Record Sheet (QRS) shall be completed as described in A-P9-000-015/PT-000 and distributed to the respective MARLANT/MARPAC Headquarters for action; and

(3) The OJPR shall be retained on the individual's UER until update action is reflected on the Personal Record Resume.

PROGESS REVIEW

6. Progress review shall be carried out quarterly by the section head (NCMs) or Training Officer (Officers).

PERIOD	OJPR REVIEWED BY (RANK, POSITION, NAME)	SIGNATURE	DATE
Week 2			
Week 4			
Week 6			
Week 8			
Week 10			
Week 12			

TRAINING COMPLETION CERTIFICATE

Date training completed: _____

_____

[Section Head (NCMs) OR HOD (Officers) Title] Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed training to the standards stated in QSP-2OOW, except as noted, and has successfully completed on the job training and has successfully completed the Fleet Examination Board. The member is qualified to carry out the duties of Second Officer of the Watch and is [NQual - ALDE] qualified.

Unit CO Date

ON-JOB PERFORMANCE RECORD SHEET

7. The remarks column in this annex shall be used for the following information only:

a. The unit in which the training and assessment were carried out; and

b. Any unusual circumstances which preclude compliance with an objective.

PO	PERFORMANCE STATEMENT	DATE	SIGNATURE	REMARKS (INCLUDE UNIT)
PO 001	Supervise Ship Control (SCOOW)			
PO 002	Conduct Dived Operations			
PO 003	Conduct Periscope Watchkeeping			

USE OF OJPR

8. This OJPR consolidates and upgrades the training received on the Second Officer of the Watch Course. Completion of the training requirements outlined in this publication as well as successful completion of the formal course will certify that the trainee has achieved the standard required for the Second Officer of the Watch MITE code [120231/120232].

9. HMC Submarines shall use this OJPR as the primary document governing the content, organization and conduct of the Second Officer of the Watch OJT. Summarized herein is the pertinent information, data, and documents required by training officers and supervisors when organizing and conducting OJT.

10. In addition, HMC Submarines shall use this OJPR:

a. To produce detailed training programs which will ensure that all trainees are afforded the opportunity to consolidate training received; and

b. As a control document to ensure that the trainee has, in fact, completed training to the standard stated in the POs as detailed in Chapter 2 of the Second Officer of the Watch QSP.

11. On completion of training, a copy of the OJPR (Annex E) and Certificate of Qualification shall be placed on the trainee's file. The OJPR shall be made out as detailed in A-PM-245-001.FM-A03, Personnel Management Information System.

OJPR ADMINISTRATION

12. This OJPR can be progressed concurrently with other OJPRs, to include:

a. Marine Systems Engineering Head of Department Qualification (NQual AISE, MITE 114809); and

b. Naval Combat Systems Engineering Head of Department Qualification (NQual AISF, MITE 114808).

13. This document will be distributed to the trainee upon successful completion of the Second Officer of the Watch residential course conducted at CFNOS.

14. Canadian Forces Naval Operations School will be responsible for issuing this package to trainees prior to leaving the school. The Submarine Training Division will ensure that each OJPR contains an up to date electronic version of the Second Officer of the Watch QSP.

PREREQUISITES

15. Before undertaking this OJPR, the trainee must have successfully completed the residential Second Officer of the Watch course conducted at CFNOS.

RELATED DOCUMENTS

16. The documents related to this OJPR are:

a. QSP-SUB-2OOW, and

b. A-PD-055-003/PQ-001 SS AILV.

QUALIFICATION

17. To be awarded the Second Officer of the Watch Qualification, the trainee must:

a. Meet all crse prerequisites as per Chapter 1 of this QSP;

b. Successfully complete the formal Second Officer of the Watch Course at CFNOS;

c. Successfully attain signatures for all training requirements contained within this OJPR package within 84 days (12 weeks) at sea; and

d. Successfully challenge the Second Officer of the Watch board.

18. The OJPR sheet contains all the OJPR POs. When the qualification board is successfully challenged, the "date training completed" section is to be filled out by the Training Officer. On successful completion of the qualification board, one copy shall be placed on the member's Unit Employment Record (UER) and CFNOS Training Division be informed via message recommending Second Officer of the Watch OJPR MITE Code [120232] be granted.

QUALIFICATION BOARD

19. Trainees shall sit and successfully challenge a qualification board before being awarded the Second Officer of the Watch Qualification. This board is the final requirement prior to CFNOS granting the ALDE National Qualification. Preferably, this board shall consist of personnel from outside the unit the OJPR was completed on.

20. A Submarine Command Qualified Officer shall chair this board and examine the trainee on all aspects of material covered during the Second Officer of the Watch training. As a minimum, the board shall consist of:

a. A Submarine Command Qualified Officer (AEDM);

b. A Submarine Qualified MARS HOD (CBTO or XO) (AEDN or AILU);

c. A Submarine Qualified MSEO (AILV, AISE);

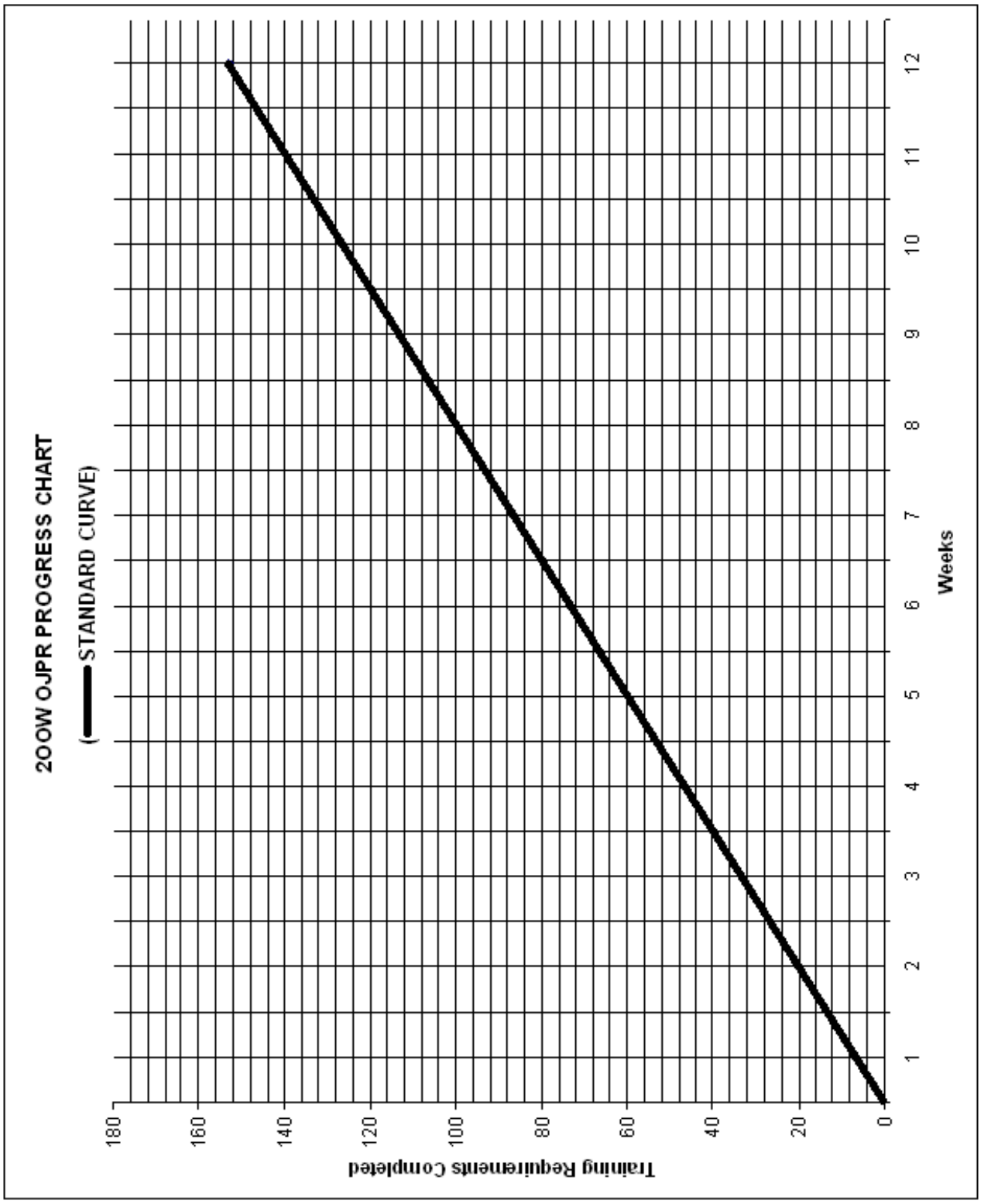
d. A Submarine Qualified CSEO (AILV, AISF); and

Should the board be conducted alongside in Halifax, a CFNOS Submarine QA or qualified Submarine Division member must be present in the role of observer to ensure standardized assessment.

UNSATISFACTORY COURSE PROGRESS

21. IAW MARCORD 9-60.

2OOW OJPR PROGRESS CHART



TRAINING REQUIREMENT SIGNATURE RECORD

*Note only designated OPI authorized to sign for these training requirements

EO 001.01 MAINTAIN TRIM				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe Depth change as SCOOW, to include: - Trimming Principles - Impact of Speed and Depth	MSEO			
Supervise Depth Change as SCOOW	MSEO			
Catch a Stopped Trim	MSEO			
Describe OMC Operation, to include: - Architecture - Modes of Operation - Alarms - Available Instrumentation - Interpretation of Readings	MSEO			
Supervise OMC Operations	MSEO			
Utilize the MLD	OOW			
Supervise Operation of Control Surfaces, to include: - Principles - Planing and Steering Techniques	MSEO			
Supervise Operation of Control Surfaces	MSEO			
React to Environmental Changes, to include: - Rough Weather - Bathymetry	MSEO			
Describe the Impact of Bulk Liquids on the Trim, to include: - Types of Bulk Liquids - Concept of Specific Gravity - Use of the Trim Crib - Impact of Standard Evolutions on Trim	MSEO			

EO 001.02 CONDUCT DIVED SOP'S AS SCOOW				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe snorting, to include: - Procedure - Stop Snorting Parameters - Breaking the Vacuum - Reports - Stop Snorting Procedures	MSEO			
Conduct a Snort	MSEO			
Describe Battery charging operations, to include: - Interdependency of Engine Loading, Battery - Charging Rate and Submarine Speed - Floating the Load - Setting Maximum Revolutions to Float the Load - Trickle Charging - Battery Charging Parameters - Reports	MSEO			
Supervise Battery Charging Operations	MSEO			
Outline the sequence of events required to dive the submarine, to include: - Procedures - Open Up for Dive (All Compartments) - Diving Stations - Long and short post diving checks	MSEO			
Define differences between Watch Diving and Pre-diving checks	MSEO			
Dive the Submarine	MSEO			
Describe the sequence of events required to surface the submarine, to include: - Methods - Preparations - Procedures - Falling Out to a Surface Routine	MSEO			
Surface the Submarine	MSEO			

EO 001.02 CONDUCT DIVED SOP'S AS SCOOW				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe the sequence of events required to return tom periscope depth as SCOOW, to include: - Preparations - Equipment Line-ups - Personnel - Reports - Procedures	MSEO			
Return to Periscope Depth as a SCOOW	OOW			
Supervise BWA operations, to include: - Personnel - Deployment and Recovery Procedures - Manoeuvring Limitations	MSEO			
Describe Domestic routines, to include: - Blowing Slop Drain and Sewage - Ditching Gash - Reports - Record Keeping - Limitations - Environmental Considerations	MSEO			
Conduct Domestic Routines	MSEO			
Manage Bilge Water, to include: - Pumping Bilges - Reports - Record Keeping - Limitations - Environmental Considerations	MSEO			
Supervise the Burning of O2 Candles as SCOOW	MSEO			
Supervise the Operation of the CO2 Absorption Unit	MSEO			
Describe the Pumping of Battery sumps, to include: - Safety considerations - Procedure	MSEO			

EO 001.02 CONDUCT DIVED SOP'S AS SCOOW				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Supervise the Pumping of Battery Sumps	MSEO			
Describe Shutting Off/Opening Up from Attack and Counter Attack	MSEO			
Conduct an Air Charge	MSEO			
Supervise the Operation of the 5-Man Chamber	MSEO			
Describe SSE Operations, to include: - Methods - Stores - Capabilities and Limitations - Procedures - Reports	CSEO			
Supervise SSE Operations, to include: - 2039 Bathy Operations	CSEO			
Describe WHDS Operation, to include: - Tube Rear Door Procedures - TOC Operation - HCP Operation - Water Shot Procedure - Weapons' Discharge Envelope - Ship Control Considerations for Weapons Employment	CSEO			
Supervise WHDS Operation	CSEO			
Describe main propulsion system, to include: - Architecture - Control System - Modes of Control (Primary, Secondary, Man Modes 1 & 2) - Modes of Operation (SSMG, GD, GU, BIS) - Capabilities and Limitations - Casualty Modes of Operation	MSEO			
Supervise Operation of the Main Propulsion System	MSEO			

EO 001.02 CONDUCT DIVED SOP'S AS SCOOW				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Utilize the Propulsion Power Statement, to include: - Limitations - Normal and Emergency Orders - Captain's Rules - Reports	MSEO			
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of Air Systems	MSEO			
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of Ventilation Systems	MSEO			
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of Cooling Systems (Sea Water, Fresh Water, Chilled Water)	MSEO			
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of Trim, Bilge and Ballast Systems	MSEO			
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of Electrical Systems	MSEO			
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of Hydraulics	MSEO			
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of Planes and Steering	MSEO			

EO 001.02 CONDUCT DIVED SOP'S AS SCOOW				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe Normal and Abnormal Modes of Operation, Equipment Line-ups, System Limitations in Alternate Modes and Consequences of Failure of the Fuel System	MSEO			

EO 001.03 CONDUCT DIVED EOP'S AS SCOOW				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe required actions for Emergency Stations	MSEO			
Describe Initial Ship Control Actions and Consequences for a Collision	MSEO			
Describe Initial Ship Control Actions and Consequences for a Grounding	MSEO			
Describe Initial Ship Control Actions and Consequences for an Abandon Ship	MSEO			
Describe Initial Ship Control Actions and Consequences for a Flood	MSEO			
Describe Initial Ship Control Actions and Consequences for Control Surface Failures	MSEO			
Describe Initial Ship Control Actions and Consequences for a Man Overboard	MSEO			
Describe Initial Ship Control Actions and Consequences for a Hydraulic Burst	MSEO			
Describe Initial Ship Control Actions and Consequences for an AC Electrical Failure	MSEO			
Describe Initial Ship Control Actions and Consequences for a DC Electrical Failure	MSEO			
Describe Initial Ship Control Actions and Consequences for a Fire	MSEO			
Describe Initial Ship Control Actions and Consequences for a Fire in the Battery Switchboard	MSEO			
Describe Initial Ship Control Actions and Consequences for a Fire in a Main Battery	MSEO			
Describe Initial Ship Control Actions and Consequences for a Fire in the Main Propulsion Switchboard	MSEO			

EO 001.03 CONDUCT DIVED EOP'S AS SCOOW				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe Initial Ship Control Actions and Consequences for Propulsion Failures	MSEO			
Describe Initial Ship Control Actions and Consequences for a Fire in an Oxygen Generator	MSEO			
Describe Initial Ship Control Actions and Consequences for a Pyrotechnic Activation	MSEO			
Describe Initial Ship Control Actions and Consequences for a HAZMAT Spill	CSEO			
Describe Initial Ship Control Actions and Consequences for a Communications Failure	CSEO			
Describe Initial Ship Control Actions and Consequences for a Failure of Engine to Stop	MSEO			
Describe Initial Ship Control Actions and Consequences for a Casualty	MSEO			
Conduct Smoke Clearance	MSEO			
Successfully conduct EOPs as SCOOW	CO			
Describe DC incident turnover, to include: - Turn Over Prerequisites - Duties and Responsibilities - Communications during Emergency Stations (DC Net, VICES, VCS 446)	MSEO			
Turn Over Damage Control Incident to HQ1	MSEO			
Take Follow Up Actions to EOPs, to include: - Briefing Command - System Restorations	MSEO			
Fall Out Emergency Stations, to include: - Prerequisites - Pipe	MSEO			

EO 002.01 INTERPRET SUB SENSOR DATA				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe ESM Equipment, to include: - Frequency Coverage - Bearing Accuracy - Architecture	CSEO			
Conduct Warner Procedures, to include: - Warner Look - Warner Clearance - Warner Take - Racket Dangerous - Operator Reports	OOW			
Describe Radar Equipment, to include: - Capabilities and Limitations - Architecture - Frequencies - SDM Interface	CSEO			
Conduct a Radar Cut Procedure	OOW			
Describe 2040 Sonar Equipment, to include: - Capabilities and Limitations - Architecture (Passive, Active and Intercept) - Frequencies - Modes of Operation (Normal, Fallback, etc.) - Automatic and Manual Tracking - Operator Reports	CSEO			
Describe PRS Sonar Equipment, to include: - Capabilities and Limitations - Architecture - Frequencies - Modes of Operation - Automatic and Manual Tracking - Operator Reports - Passive Ranging Principles	CSEO			
Describe 2046 Sonar Equipment, to include: - Capabilities and Limitations - Architecture (Flank, Towed Array) - Frequencies - Modes of Operation - Automatic and Manual Tracking - Operator Reports	CSEO			

EO 002.01 INTERPRET SUB SENSOR DATA				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe 189 Cavitation Indicator	CSEO			
Describe 2004 Bathy	CSEO			
Describe 2039 Bathy, to include: - SSXBT - SSXVT	CSEO			
Supervise Sweeps for Clearance	OOW			
Supervise a Stern Arcs Clearance	OOW			
Describe Automatic Information System (AIS) to include: - Capabilities and Limitations - Operation	CSEO			

EO 002.02 CONDUCT BROADBAND TARGET MOTION ANALYSIS				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Explain Broadband TMA, to include: - Broadband Theory - Opening/Closing Curves - Single Leg Eklund - Double Leg Eklund - Application of Ballpark Range - PBOOT - Solution at CPA	OOW			
Conduct Manual TMA	OOW			
Describe CCS 876, to include: - Architecture - Capabilities and Limitations - Operation and Functions - WSDB Interface - Boot/Reboot	CSEO			
Operate the CCS 876	CSEO			
Conduct Track Management on CCS 876, to include: - Page Management - Contact Reports - RMS - Updating Solutions - Zig Procedure	OOW			
Conduct a Ranging Manoeuvre on CCS 876, to include: - PBOOT - Reports - Ballpark Range	OOW			
Conduct Visual Set-Up on CCS 876	OOW			
Supervise a Snap Firing	OOW			

EO 002.02 CONDUCT BROADBAND TARGET MOTION ANALYSIS				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Act as LOPO, to include: - Setting Up the Plot - Target Set Ups - Speed between the looks - Maximum/Minimum Range for Speed - Crossover range - Determining Contact Solutions - Zig Procedures	NAVO			

EO 002.03 CONDUCT NARROWBAND TARGET MOTION ANALYSIS				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Explain Narrowband TMA, to include: - Narrowband Theory - Characteristics affecting the Towed Array (TA) - Resolving for Bearing Ambiguity - Triangulation Broadband Ranging - Narrowband Definitions - Doppler Effect - Frequency Relationships - Frequency Calculations - Frequency Leg Eklund - BFP Plotting Techniques - CCS 876 Towed Array Function	OOW			
Operate the CCS 876 Towed Array Function	OOW			

EO 002.04 EMPLOY EXTERNAL COMMUNICATION SYSTEMS				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Outline RF Communications, to include: - VHF Operation - Remote Circuits in Control - Non-Secure vs. Secure Circuits - NUCCO/Authentication Procedures - Voice Procedures - Brevity Codewords - Daily Changing Call Signs	RS			
Communicate Using RF Communications	RS			
Describe Radio Equipment, to include: - Fitted VHF - UHF/HF Transceivers - LF/VLF Receivers - Modems - Frequency Bands - Cryptographic Equipment - Antennaes - Capabilities and Limitations - ECBs - Meteorological Computer - System Line-ups - GCCS	CSEO			
Describe Submarine Broadcast Procedures, to include: - Routines - Broadcast Control Authority Structure and Organization - Message Handling - Submarine Accountability Messages - Operator Reports	OOW			
Describe 2008 UWT, to include: - Transducers - Frequency Bands - Operating Modes - UWT Codes - Ranging Methods - HOOKER Procedure - UNCLE JOE Procedure - Scanner Reports	CSEO			

EO 002.04 EMPLOY EXTERNAL COMMUNICATION SYSTEMS				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Operate 2008 UWT	UC1			
Describe Operational Messages, to include: - SUBNOTE to include: - SUBNOTE Req; - SUBNOTE chg; and - SUBNOTE Chg Req Diving Message; RSS/ESR; CASEX to include: Starred relaxation req. Daily SITREP; MARINTREP; BARNSTORM; OPSTAT Unit; and RFI/RFS.	OOW			

EO 002.05 COORDINATE THE WATCH				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Utilize operational references to extract pertinent information, to include: - CASEX - BARNSTORMs - OPAREA Assignments - FLEX Programme - MARLANTORD 44-4 - OPSKED - ATP 1 Vol1 - ATP 28 - AXP 1 - AXP 5 - ATP18 - APP4 - APP7 - APP11 - STOPS - CFCD 110 - SMP 108	OOW			
Direct At-Sea Watch Routines, to include: - Dived Routines - Surface Routines	XO			
Conduct Watch Turnover, to include: - Pre-Watch Rounds - Post-Watch Rounds - Captain's Night Orders - Navigation - Sensors - Contact Situation - Ship Control - Equipment Considerations - Watch Brief	OOW			
Conduct Daily User Checks	OOW			
Maintain Records, to include: - Operational Records - Logs	XO			

EO 002.05 COORDINATE THE WATCH				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Describe Signature Management, to include: - Organization - Noise Rounds - Purpose - Quiet States - Noise Husbandry	CSEO			
Enforce Signature Management Policies	CSEO			
Describe CIO, to include: - Personnel Required - Duties and Responsibilities - Information Flows	XO			
Describe Atmospheric Monitoring, to include: - Routine Monitoring - Actions to be taken in event of Degraded Atmosphere	OOW			
Supervise Atmospheric Monitoring	OOW			
Conduct a Close Quarters Drill, to include: - References - Manoeuvring	OOW			
Conduct Fishing Vessel Avoidance, to include: - References - Manoeuvring	OOW			
Describe Environmental Regulations	MSEO			

EO 002.06 CALCULATE SUBMARINE POSITION				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Define POE, to include: - Principles - Electronic POE Theory - Implications of POE intersecting Navigation Hazards - Sources of Error	NAVO			
Input POE into SDM	NAVO			
Interpret POE	NAVO			
Describe POE reduction, to include: - Methods - Limitations	NAVO			
Reduce POE	NAVO			
Describe WSM Techniques, to include: - Rationale - Interpretation - Types - SDM Functions	NAVO			
Apply WSM techniques	NAVO			
Describe PMI Techniques, to include : - Rationale - Message Interpretation (NOI, SUBNOTE, DIVER) - Types (MHN, OPAREAs) - SDM Functions	NAVO			
Apply PMI techniques	NAVO			

EO 002.06 CALCULATE SUBMARINE POSITION				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Explain Safety Separations, to include: - BVSS - UVSS - Safe Depth - Use of AXP1	OOW			
Calculate Safety Separations, to include: - BVSS - UVSS - Safe Depth - Use of AXP1	OOW			
Utilize Dived Charts	NAVO			
Recognize significance of colours	NAVO			
Describe how assigned areas are displayed on dived charts	NAVO			
Interpret data displayed on the dived chart	NAVO			
Conduct Bottom Contour Navigation	NAVO			
Define SDM to include: - Architecture - Chart Types (Raster/Vector, etc.) - Capabilities and Limitations - Speed Times Distance Functions - Information Sources - GPS - POE - Inputting Visual and Radar Fixes - Route Functions - AIS Functions - Datums	NAVO			
Operate SDM	NAVO			

EO 002.06 CALCULATE SUBMARINE POSITION				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Define Echo sounder to include: - Fitted Equipment - Frequencies - Capabilities and Limitations - Snap Sounding - Sounding Matrix (NAVO)	UC1			
Operate Echo Sounder	UC1			

EO 003.01 MAINTAIN SUB SAFETY AT PERISCOPE DEPTH				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Maintain a Periscope Watch (Have the Look)	CO			
Define Periscope (Search and Attack) to include: - Capabilities and Limitations - Construction - Operation - Knobology - Settings - TI	CSEO			
Operate Periscopes (Search and Attack)	CSEO			
Conduct an Emergency Go Deep as a Periscope Watch Keeper	CO			
Conduct a Visual Set Up	OOW			
Conduct a Periscope Watch Turnover	OOW			
Make a Visual Contact Report	OOW			
Operate the TTVC	CSEO			
Describe Periscope Imagery, to include: - Equipment - Settings - Operation	CSEO			
Conduct Periscope Imagery	CSEO			
Describe Periscope Watch Keeping Techniques, to include: - ARL Routines - Sinuation - Mast Exposure Limits - Approach Limitations - Mast States - Danger Levels	OOW			

EO 003.01 MAINTAIN SUB SAFETY AT PERISCOPE DEPTH				
Performance Requirements	**OPI**	**Signature**	**Date**	**Comments**
Calculate DOT of Contacts	OOW			
Describe Go Deep Circles, to include: - Theory - Construction - Application - Range Rates	OOW			
Describe operation of Masts, to include: - Mast Speed Limitations - Modes of Operation (Local, Remote, etc.)	MSEO			
Supervise Operation of Masts	MSEO			

ANNEX F – ACRONYMS AND ABBREVIATIONS

ACRONYM OR ABBREVIATION	TERM OR DEFINITION
AFHPDW	AWAY FROM HOME PORT DUTY WATCH
ATB	ANGLE ON THE BOW
BR	BOOK OF REFERENCE
BSQ	BASIC SUBMARINE QUALIFICATION
BWA	BUOYANT WIRE AERIAL
CASEX	COORDINATED ANTI-SUBMARINE EXERCISE
CCA	CURRICULUM CONTROL AUTHORITY
CFCD	CANADIAN FORCES CLASSIFIED DOCUMENT
CFITES	CANADIAN FORCES INDIVIDUAL TRAINING AND EDUCATION SYSTEM
CFNOS	CANADIAN FORCES NAVAL OPERATION SCHOOL
CIO	COMBAT INFORMATION ORGANIZATION
CO	COMMANDING OFFICER
COTW	CHIEF OF THE WATCH
CPA	CLOSEST POINT OF APPROACH
CRCN	COMMANDER ROYAL CANADIAN NAVY
CSO	CAPTAIN'S STANDING ORDERS
CSRM	CANADIAN SUBMARINE RECORD'S MANUAL
CSSOM	CANADIAN SUBMARINE SONAR OPERATOR'S MANUAL
DC	DAMAGE CONTROL
DLE	DOUBLE LEG EKLUND
DNTE	DIRECTOR NAVAL TRAINING AND EDUCATION
DOA	DURATION OF ATTACK
DOT	DISTANCE OFF TRACK
DSMST	DUTY SENIOR MARINE SYSTEMS TECHNICIAN
EBS	EMERGENCY BREATHING SYSTEM
EC	ENABLING CHECK
ECB	EXPENDABLE COMMUNICATIONS BUOY
ECPINS	ELECTRONIC CHART PRECISE INTEGRATED NAVIGATION SYSTEM
EMS	ENVIRONMENTAL MANAGEMENT SYSTEM

EO	ENABLING OBJECTIVE
EOP	EMERGENCY OPERATING PROCEDURE
FIV	FIRE ISOLATION VALVE
FLTSOP	FLEET STANDARD OPERATING PROCEDURES
FMFCS	FLEET MAINTENANCE FACILITY CAPE SCOTT
GIV	GROUP ISOLATION VALVE
HAZMAT	HAZARDOUS MATERIAL
HCI	HARBOUR CONTROL INSTRUCTIONS
HOD	HEAD OF DEPARTMENT
HP	HIGH PRESSURE
HPA	HIGH PRESSURE AIR
HPDW	HOME PORT DUTY WATCH
HQ	HEADQUARTERS
IAW	IN ACCORDANCE WITH
KYD	KILO YARD
LOP	LOCAL OPERATING PLOT
LOPO	LOCAL OPERATING PLOT OFFICER
LS	LESSON SPECIFICATION
MLD	MANOEUVRING LIMITATIONS DIAGRAM
NCPM	NAVAL COMBAT PROCEDURES MANUAL
NEM	NAVAL ENGINEERING MANUAL
NTI	NAVAL TRAINING INSTRUCTION
NTO	NAVAL TECHNICAL OFFICER
ODS	OZONE DEPLETING SUBSTANCE
OJPR	ON JOB PERFORMANCE RECORD
OMC	ONE MAN CONTROL
OOD	OFFICER OF THE DAY
OOW	OFFICER OF THE WATCH
OPAREA	OPERATION AREA
PBOOT	PREDICTED BEARING RATE OUT OF TURN
PC	PERFORMANCE CHECK
PMI	PREVENTION OF MUTUAL INTERFERENCE
PO	PERFORMANCE OBJECTIVE

POE	**POOL OF ERRORS**
QSP	**QUALITY STANDARD AND PLAN**
RF	**RADIO FREQUENCY**
RMS	**RELATIVE MOTION SOLUTION**
SCOOW	**SHIP CONTROL OFFICER OF THE WATCH**
SCT	**SHIP CONTROL TRAINER**
SDM	**SHINNADS DUAL MILITARY**
SFCS	**SUBMARINE FIRE CONTROL SYSTEM**
SHINNADS	**SHIPBOARD INTEGRATED NAVIGATION AND DISPLAY SYSTEM**
SITREP	**SITUATION REPORT**
SLE	**SINGLE LEG EKLUND**
SWD	**SUBMARINE WARFARE DIRECTOR**
SO	**STANDING ORDERS**
SOMADS	**SUBMARINE ORDNANCE MOVEMENT AND DISCHARGE**
SOP	**STANDARD OPERATING PROCEDURES**
SOQC	**SUBMARINE OFFICER QUALIFICATION COURSE**
SSE	**SUBMARINE SIGNAL EJECTOR**
SSTG	**SUBMARINE SEA TRAINING GROUP**
STOPS	**SUBMARINE TACTICAL OPERATING PROCEDURES**
SUBSAR	**SUBMARINE SEARCH AND RESCUE**
TAT	**TRAINING ADMINISTRATION TIME**
TBP	**TO BE PROPAGATED**
TMA	**TRACK MOTION ANALYSIS**
TOC	**TUBE OPERATORS CONSOLE**
TP	**TEACHING POINT**
TWOPS	**TACTICAL WEAPONS SYSTEM OPERATING PROCEDURES**
UWT	**UNDER WATER TELEPHONE**
VCEOP	**VICTORIA CLASS EMERGENCY OPERATING PROCEDURES**
VCS	**VICTORIA CLASS SUBMARINE**
VCSSO	**VICTORIA CLASS STANDARD OPERATING PROCEDURES**
VSCTT	**VICTORIA SUBMARINE COMMAND TEAM TRAINER**
WHDS	**WEAPONS HANDLING AND DISCHARGE SYSTEM**

WSM	**WATER SPACE MANAGEMENT**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

PO/EO	PO / EO TITLE	EO (x50 min periods)	EC* (x50 min periods)	PC* (x50 min periods)	Total (x50 min periods)
EO 001.01	Maintain Trim	18	2		
EO 001.02	Conduct SOPs as SCOOW	56	3		
EO 001.03	Conduct Dived EOPs as SCOOW	84	16 (2 per trainee)		
PO 001	**Supervise Ship Control**	**158**	**21**	**16 (2 per trainee)**	**195**
EO 002.01	Interpret Sensor Data	14	2		
EO 002.02	Conduct Broadband Target Motion Analysis	39	2		
EO 002.03	Conduct Narrowband Target Motion Analysis	38	2		
EO 002.04	Employ External Communication System	17	2		
EO 002.05	Coordinate the Watch	12	2		
EO 002.06	Calculate Submarine Position	32	3		
PO 002	**Conduct Dived Operations**	**152**	**13**	**8 (1 per trainee)**	**173**
EO 003.01	Conduct Periscope Watchkeeping	72	4		
PO 003	**Conduct Periscope Watchkeeping**	**72**	**4**	**8 (1 per trainee)**	**84**
SECOND OFFICER OF THE WATCH COURSE TOTAL		**382**	**38**	**27**	**452**

* Includes review and debrief time.

ANNEX H – ADDITIONAL TRAINING RESOURCES

In accordance with Chapter 1 paragraph 3 add any extra tables or paragraphs used for Resources in this annex. (**Not** to be deleted.)

ANNEX I – COSTING INFORMATION

Include information about cost in whichever format is applicable.

This annex is optional (**Not** to be deleted).

ANNEX J – COURSE HISTORY

This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reason for amendment is to be provided here.

1. The COTW course was initially developed and piloted. After doing so, it was decided to merge the AJSL – Chief of the Watch (COTW) course and the AILV – Sub Officer Qualification Course in order to create the Second Officer of the Watch (2OOW) course, to better align with the job. The Navigation portion from the COTW crse has been moved into a new PO in the FNO course taught on the west coast. SPQR paperwork to create the new SS will be submitted to D Nav P & T in conjunction with the QSP.

2. 17 Dec 15 – Amended QSP to include NQual ALDE. Completed by NPTG Quality Control.

ANNEX K – ASSESSMENT PLAN

Note: This annex is optional and **may be** deleted.

EC/PC	Scope Covered	How	Time (x50 min Periods)	Pass/Fail Criteria	Resources Required	Limitations
EC 001.01	EO 001.01	Practical	2 (per trainee	Pass (EC Checklist)	SCT, References	SCT Availability
EC 001.02	EO 001.02	Written	3	80%	Classroom	Nil
EC 001.03	EO 001.03	Practical	2 (per trainee)	Pass (EC Checklist)	SCT, References	SCT Availability
PC 001	PO 001	Practical	2 (per trainee)	Pass (PC Checklist)	SCT, References	SCT Availability
EC 002.01	EO 002.01	Written	2	80%	Classroom	Nil
EC 002.02	EO 002.02	Written	2	80%	Classroom	Nil
EC 002.03	EO 002.03	Written	2	80%	Classroom	Nil
EC 002.04	EO 002.04	Written	2	80%	Classroom	Nil
EC 002.05	EO 002.05	Written	2	80%	Classroom	Nil
EC 002.06	EO 002.06	Written	2	80%	Classroom	Nil
PC 002	PO 002	Practical	8 (per trainee)	Pass (PC Checklist)	VSCTT	VSCTT Availability, Restricted Access Area
EC 003.01	EO 003.01	Practical	4 (per trainee)	Pass (EC Checklist)	VSCTT	VSCTT Availability, Restricted Access Area
PC 003	PO 003	Practical	7 (per trainee)	Pass (PC Checklist)	VCSTT	VSCTT Availability, Restricted Access Area